

RIJUSMIT BISWAS

New Delhi, India | rijusmit.biswas@gmail.com | +91 8929693930 | linkedin.com/in/rijusmit-biswas | github.com/riju-talk | rijusmit.vercel.app

SUMMARY

CS senior at IIIT Delhi, graduating September 2026. I ship software end to end and stay with it after it goes live. I have taken a SaaS product from an idea to a running deployment on AWS, led a stack migration on a live system without any downtime for users, and built a portal that over 500 students use, on a schema and API contracts I designed for other engineers to build against. I also build the parts around the code that keep it working, including Dockerised services, CI that runs tests and static analysis on every pull request, retry and rate limit handling, and structured logging. I have put machine learning into production as well, including a semantic evaluation pipeline that my lab still uses.

EDUCATION

B.Tech, Computer Science and Engineering, IIIT Delhi | Operating Systems, Networks, Distributed Systems, DBMS, DSA **Nov 2022 to Sept 2026**

EXPERIENCE

Production Delivery and a Live Stack Migration, Software Engineering Intern, Shipzee **March 2026 to Aug 2026**

- I shipped features across a live courier aggregation SaaS, from the React screens through the API layer down to the database, working the full development cycle alongside internal teams and outside partners.
- I led a stack migration to make the system easier to scale and maintain. I sequenced the cutover so that users saw no downtime at all.
- I traced production failures across service boundaries using logs and request traces, then verified the fixes against live traffic. I also wrote unit and integration tests over third party integrations, which caught contract mismatches before release.

A Student Services Portal at Institutional Scale, Head of Product Development, IIIT Delhi Student Council **Aug 2025 to Present**

- I built and shipped a portal that over 500 students use, with React on the front and Node.js, Express and PostgreSQL behind it. I owned the schema design, the indexing, and the API contracts that the other engineers built against.
- I set up GitHub Actions for CI and CD with automated tests, and I ran code review for 8 engineers. I wrote the API contracts down so that parallel work merged cleanly.
- I replaced manual data entry with validated forms. That cut administrative work by 60 percent and removed a recurring class of data integrity bugs.

A Semantic Evaluation Pipeline for Model Outputs, AI Engineer and UG Researcher, MIDAS Lab, IIIT Delhi **May 2025 to Jan 2026**

- I built HorusLLM, a semantic evaluation pipeline that scores model generations along several alignment dimensions and flags degraded outputs on its own, in place of manual spot checking. The auditing stage cut hallucinations by 20 percent. My lab still uses it, the harness and benchmarks are open source, and I am first author on a paper about it now under review at AAAI 2027.
- I ran reproducible pipelines over three datasets, with retry logic and rate limit handling for outside APIs, so that a rerun produced the same artifacts every time.

A Geospatial Pipeline for Agricultural Prediction, ML Research Intern, CSS Lab, IIIT Delhi **Jan 2025 to May 2025**

- I built a satellite imagery pipeline on Google Earth Engine with parallel preprocessing, producing segmentation masks and multi channel inputs at district scale. I measured how far the data drifted across more than 10 districts, so that accuracy was established outside the training distribution rather than on one held out split.

PROJECTS

Hostiggo, Production SaaS Platform | Node.js, React, PostgreSQL, AWS, Docker, GitHub Actions

- I took a product from an idea to a running cloud deployment. I owned the React front end, the whole backend, the database design and the infrastructure, and I kept it running in production.
- I built CI and CD with monitoring from the first day, so deploys stayed reliable as usage grew instead of being patched in later under pressure.

DreamDOT, Distributed GraphQL Platform | GraphQL, Node.js, Microservices, MongoDB, Docker

- I designed a distributed backend with GraphQL federation across several services, resolving queries that cross services without falling into N+1 problems.
- I defined the service boundaries and the schema contracts, so that teams could ship on their own without breaking the unified graph.

Flourish, AI Agent Service | FastAPI, LangChain, OpenAI, Firebase, Docker, GitHub Actions

- A FastAPI backend that runs multi step LLM tool calls against outside APIs, with Firebase Auth and Firestore for identity and storage. Retry logic and graceful degradation keep a failure upstream from breaking the user flow.

MineMEETS, Multimodal Search Pipeline | CLIP, Whisper, Pinecone, Docker, GitHub Actions, Pytest

- Ingestion pipelines that generate visual, text and audio embeddings behind a single interface, with dimension checks and reprocessing that is safe to repeat, so a failed run can simply be run again. Pinecone namespace isolation makes reindexing and rollback safe, and CI runs Pytest, Black, Pylint and MyPy on every pull request.

TECHNICAL SKILLS

Backend and APIs: Python (FastAPI), Node.js, Express.js, REST and GraphQL API design, GraphQL federation, Socket.IO, authentication, microservices, idempotent processing, retry and rate limit handling

Frontend: React.js, Next.js, Tailwind CSS, responsive layouts, dashboards, REST and GraphQL clients, WebSocket UIs

Infrastructure and Delivery: AWS, Docker, GitHub Actions for CI and CD, Linux, shell scripting, Vercel, Pytest, unit and integration testing, static analysis with Black, Pylint and MyPy, structured logging, code review

Data and Distributed Systems: PostgreSQL, MongoDB, Prisma ORM, Firestore, schema design, indexing, query optimisation, PySpark, Airflow, ETL pipelines

AI, ML and Languages: OpenAI APIs, LangChain, agentic tool calling, RAG with Pinecone, embeddings with CLIP and Whisper, PyTorch, HuggingFace, LoRA and QLoRA, Google Earth Engine, and Python, JavaScript, TypeScript, C++, SQL and C

OPEN SOURCE, ALGORITHMS AND LEADERSHIP

- **Open source:** the HorusLLM evaluation harness and benchmarks, and MineMEETS, shipped with CI and tests. **Algorithms:** Codeforces Specialist at 1403, with over 2500 LeetCode problems solved. **Leadership:** Vice Chair, IEEE Student Branch.